Validity Evidence for the Spiritual Harm and Abuse Scale (SHAS)

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Chapter 1

About this Project

This is a reproducible research project for validation of the nascent Spiritual Harm and Abuse Scale (SHAS), developed by Dan Koch. It aims to function as an evolving technical report to conceptualize and document various validity studies – and, by extension, validity evidence – for this scale. We hope some of these validity considerations and empirical studies amount to meaningful publishable contributions to the nascent research literature in spiritual abuse and/or the literature in religion, counseling psychology, and/or psychological measurement.

The project is open source, for collaboration. It is built in R using the R Studio IDE [R Core Team, 2022]. No Excel, no SPSS, no Word. It is built in `bookdown` (Xie [2022]) and hosted at Github as a public repository at https://github.com/jackbhuber/SHAS. There it is served over Github Pages as a website – https://jackhuber.org/SHAS/ – for collaboration between Jack and Dan.

1.1 Research Questions

Validation is research into the meanings and uses of scores from instruments. It can be an ongoing research agenda. Following is an initial set of questions to guide research and scale improvement of the SHAS:

- **Item analysis and refinement** - Do some items have too many response categories?
- **Data reduction** - As written and executed, which items appear to be critical (best performing) and which appear to be redundant or noisy? What might a shorter, streamlined instrument look like?
- **Person fit** - How well does the current scale fit the population of examinees? Are there ceiling or floor effects? To what extent might the existing items under-represent the full range of abuse?

- **Scale invariance** - Do the scales mean the same thing to different groups of examinees? To what extent do these scales (abuse constructs) vary by age, gender, and/or sexuality?
- **Developmental pathway** - Beyond the current conceptualization of scales composed of redundant items, what might an incremental scale of abuse with diagnostic items look like?

Chapter 2

Background of the Project

2.1 Spiritual Harm and Abuse

This is where Dan can introduce and discuss spiritual harm and abuse – his history with the topic, what it might mean in real world terms, the need for scientific study of the topic, and research to date. At a minimum he can paste in text from his extant writings and/or write new text.

2.2 The Spiritual Harm and Abuse Scale (SHAS)

This might be where Dan presents the case for a clinical measure of spiritual harm and abuse and the work he has done to date to develop the SHAS. This could be a summary of his forthcoming JSSR article.

2.3 Validity Research for the SHAS

This might be where I discuss what it means to conduct validity research on any instrument, and in particular the SHAS.

Chapter 3

Content Validity

In this chapter, we consider the content validity of the Spiritual Harm and Abuse Scale (SHAS). We begin by defining content validity and discussing its importance. Then we document the work we did to map the domain of spiritual abuse for the purpose of developing items for piloting and operational use. We conclude by considering what scope exists for more work in this area that would improve future development of SHAS instrumentation.

3.1 About Content Validity

The focus of content validity is "the relationship between the content of a test and the construct it is intended to measure" [American Educational Research Association et al., 2014]. Do they match? Does the instrument contain content that is not part of the construct? Are there parts of the content unmeasured, such that a high score still underrepresents an examinee's true level of a trait?

Content validity in this case concerns the relationship between the concept of spiritual abuse and the content of the 66 items that were piloted to form the SHAS. What is spiritual abuse? What are its important parts? What counts as evidence of it? Is there more to spiritual abuse than was captured by the 66 items?

3.2 Content Specification

"Test developers often work from a specification of the content domain. The content specification carefully describes the content in detail, often with a classification of areas of content and types of items" [American Educational Research Association et al., 2014]. For the purpose of piloting items, the domain of spiritual abuse included two constructs: external (abusive) events and internal states (of emotional harm).

3.2.1 External Events

One domain of spiritual abuse is external abuse *events*. The focus is on something that happened, or a pattern of recurring behavior on the part of someone in a place of spiritual authority that was experienced by the examinee as abusive. The first 52 items of the SHAS were drawn from this domain of possible abusive events. To each item, an examinee was offered five response categories:

1. Never
2. Once or twice
3. Sometimes
4. Often
5. Always

To sum each examinee's vector of responses to these items is to accumulate evidence of abuse. In the aggregate, scores for this collection of items could range from 52 to 260. An examinee who *never* experienced *any* of these events would score 52. At the other extreme, an examinee who scored 260 had *always* experienced *all* of these events.

3.2.2 Internal States

Another important domain of spiritual abuse is *internal states*. The focus is on the subjective experience of an examinee "as a result of negative experiences in church or Christian group settings." The questionnaire included 14 items asking examinees about their emotions in response to events. To each item, an examinee was offered the same five response categories as before:

1. Never
2. Once or twice
3. Sometimes
4. Often
5. Always

To sum each examinee's vector of responses to these items is to accumulate evidence of internal emotional *harm*. Total scores for these items could range from 14 to 70. Examinees who *never* experienced *any* of these unpleasant emotions would score 14, while examinees who scored 70 had *always* experienced *all* of the difficult emotions.

3.2.3 Additional Domains

Reise and Waller [2009] make some interesting points about the nature of the traits addressed in clinical settings which provoke thought about the trait of spiritual abuse.

3.3 Summary

What counts as spiritual abuse? What counts as evidence of it?

Chapter 4

Data Collection

The data were collected online from January 28, 2021 to February 27, 2021.

A final total of 3,222 people responded to the survey at some level. Depending on how much data are missing, this is an impressively large sample size which will provide lots of power for most simple statistics and should for the most part satisfy the large-sample requirements of the advanced statistical procedures associated with measurement work (i.e., factor analysis, IRT).

However, the data come from a convenience sample which raises some challenge of sampling bias. Dan sent the link out to [lots of people], and (if memory serves) encouraged them to send it to people *they* knew. This would seem to bias responses toward people demographically similar to Dan: white, Millennial, west coast American, middle class, highly educated, primarily Protestant and Catholic Christian, etc. This will be more apparent upon closer inspection of the demographics of the sample.

One source of examinees was people who listen to progressive Christians podcasts such as *You Have Permission* which Dan produced at the time of data collection. This is a known source of bias toward people who have some interest in Christianity along with at least some level of rejection of conservative/orthodox Christianity and/or political conservatism. However, this would be an appropriate group from which to invite participation if the target population is people who have experienced spiritual abuse.

Chapter 5

Data Preparation

This chapter documents our decisions surrounding how to prepare the data for analysis. These decisions include details on loading the necessary R packages, loading the data in R, coding or recoding of variables. Also discussed is how we deal with missing data.

5.1 Preliminary work

Load R packages. Some R users load all their packages up front. Others, perhaps for instructional reasons, load them just before calling functions, perhaps for instructional reasons. I can see merit in listing all the packages for properly citing them. Here is a code chunk for loading all the packages up front:

```
# LOAD PACKAGES
library(Hmisc)  # for variable and value labels
library(naniar)  # for missing data
library(corrplot)  # for correlation plot
library(knitr)  # how RMarkdown compiles code into documents
```

Load data. The data are contained in an Excel file, simply named "data.xlsx", in the "data" folder of this project. The important thing is to keep all the data in the same folder within the larger "SHAS Validity" project folder.

Select items for analysis. Which items should we include for analysis? The first 13 columns in the original data file report data from several screening items that were placed at the beginning of the questionnaire. But because they are constant and not variable, they don't seem to have a place in scale development. Therefore it seems the important columns range from columns 14 to 97. The first set of items measure the **External Events** scale. These 52 items range from columns 14 to 35. The second set of items measure the **Internal States**

scale, and these 14 items range from columns 66 to 79. The last set of items capture **demographic** variables, and these range from columns 80 to 97.

```
# LOAD ORIGINAL DATA
library(readxl)  # for reading the source Excel file
shas <- read_excel("data/data.xlsx")  # load Excel file

# SELECT DATASETS FOR ANALYSIS
all <- shas[, 14:97]  # select items, dems from columns 14-79
all <- subset(all, select = -c(Ext_Avg:Global_Avg))  # remove columns for now
items <- all[, 1:66]
ext_items <- items[, 1:52]
int_items <- items[, 52:66]

# SAVE DATA FILES
write.csv(all, "data/all.csv", row.names = FALSE)
write.csv(items, "data/items.csv", row.names = FALSE)
write.csv(ext_items, "data/ext_items.csv", row.names = FALSE)
write.csv(int_items, "data/int_items.csv", row.names = FALSE)
```

5.2 Missing data

Missing data can be an important problem. The problem has two parts. The first is how many data are missing. Obviously the more data are missing, the larger the barrier this problem poses to analysis. The second is the mechanism of missing data. Are the missing data randomly, or is there a pattern to the missing data - like a problematic item, or a distinct group of examinees who chose not to respond? Figure 5.1 presents two plots from the VIM package for missing data [Templ et al., 2021].

```
# VISUALIZE AND QUANTIFY MISSING DATA
library(VIM)
a <- VIM::aggr(all, plot = FALSE)
plot(a, numbers = TRUE, prop = FALSE)
```

The first plot in Figure 5.1 is a frequency distribution of the number of missing data entries for each item. This plot reveals very few missing data for the first set of External Events items, but nearly 200 missing entries for the second set of Internal States items and demographics. The raises two questions. The first is: Who were those examinees? We know that they responded to the first 52 items, then apparently quit the survey. Why? Was it just fatigue, or is there more to know about them? The second question is what to do with the information they actually did provide.

The second plot in Figure 5.1 illustrates the different combinations of missing entries. The blue squares represent cases with complete responses, and the red indicates missing data. The rows represent combinations of complete and

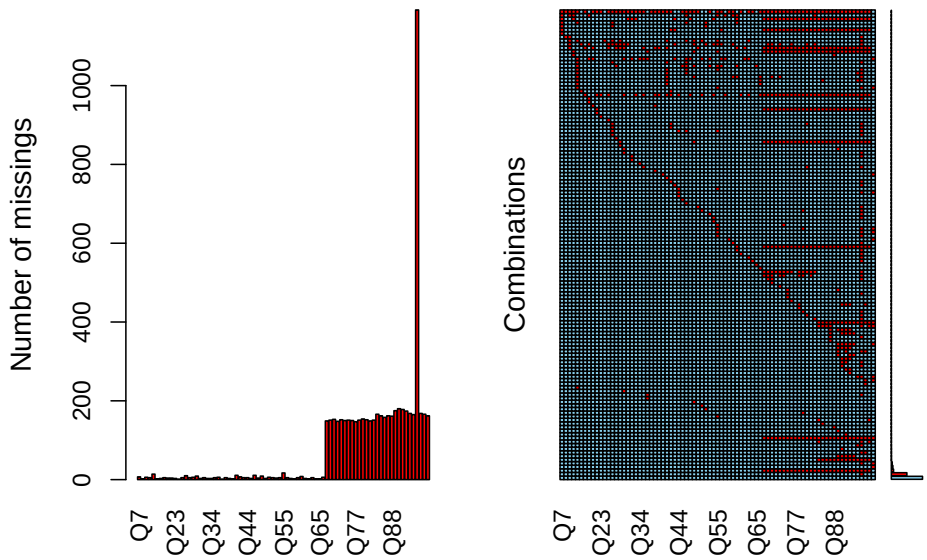


Figure 5.1: Missing Data

Table 5.1: Little's Test of Missing Completely At Random (MCAR)

statistic	df	p.value	missing.patterns
2089.341	8663	1	117

missing responses. The bottom row of entirely blue squares represents complete data – examinees who responded to all of the items. The barplot on the right axis indicates that these were the most frequent combinations. The second most frequent combination was examinees who completed all of the External Events items and then quit the survey. Beyond that, there is no apparent order or pattern to the missing data.

Mechanism of missing data. McKnight et al. [2007] prescribe several approaches for investigating the mechanisms of missing data. An appropriate first step is to test whether the data are Missing Completely At Random (MCAR) with the use of Little's test Little [1988]. This is a chi-square test of the difference between the observed pattern of missing responses and the pattern of responses that would expected if missingness were completely random. We report the results of this test in Table 5.1.

```
library(naniar)
knitr::kable(mcar_test(all), caption = "Little's Test of Missing Completely At Random (MCAR)
```

The test is fairly conclusive. The p-value for the chi-square is 1, which means observed pattern of missing data do not diverge significantly from what would

be expected if the data were missing completely at random.

Were examinees triggered? As mentioned above, a possible explanation of the missing data in the Internal States items is that examinees were triggered. As after responding to the preceding 52 items calling to mind imagery of abusive external events, these examinees were ***done*** with the survey. A possible test of this hypothesis is an examination of the relationship between a total score of responses to the External Events items and a score of total missingness on the Internal States items. The empirical work to test this hypothesis is laid out in the following code chunk. The items measuring external events are summed into a total score such that a higher value indicates more exposure to abusive external events. The items measuring internal states are coded so that 1=missing and 0=all others response, and these items are summed into a score of total missingness.

```
# 1. ISOLATE THE DATA
miss_set <- items

# 2. DUMMY CODE INTERNAL STATES ITEMS
miss_set$Q68[miss_set$Q68 > 0] <- 0
miss_set$Q69[miss_set$Q69 > 0] <- 0
miss_set$Q70[miss_set$Q70 > 0] <- 0
miss_set$Q71[miss_set$Q71 > 0] <- 0
miss_set$Q72[miss_set$Q72 > 0] <- 0
miss_set$Q73[miss_set$Q73 > 0] <- 0
miss_set$Q74[miss_set$Q74 > 0] <- 0
miss_set$Q76[miss_set$Q76 > 0] <- 0
miss_set$Q77[miss_set$Q77 > 0] <- 0
miss_set$Q78[miss_set$Q78 > 0] <- 0
miss_set$Q79[miss_set$Q79 > 0] <- 0
miss_set$Q80[miss_set$Q80 > 0] <- 0
miss_set$Q81[miss_set$Q81 > 0] <- 0
miss_set$Q82[miss_set$Q82 > 0] <- 0
miss_set$Q68[is.na(miss_set$Q68)] <- 1
miss_set$Q69[is.na(miss_set$Q69)] <- 1
miss_set$Q70[is.na(miss_set$Q70)] <- 1
miss_set$Q71[is.na(miss_set$Q71)] <- 1
miss_set$Q72[is.na(miss_set$Q72)] <- 1
miss_set$Q73[is.na(miss_set$Q73)] <- 1
miss_set$Q74[is.na(miss_set$Q74)] <- 1
miss_set$Q76[is.na(miss_set$Q76)] <- 1
miss_set$Q77[is.na(miss_set$Q77)] <- 1
miss_set$Q78[is.na(miss_set$Q78)] <- 1
miss_set$Q79[is.na(miss_set$Q79)] <- 1
miss_set$Q80[is.na(miss_set$Q80)] <- 1
miss_set$Q81[is.na(miss_set$Q81)] <- 1
```

```r
miss_set$Q82[is.na(miss_set$Q82)] <- 1

# 3. TOTAL SCORES ON EXTERNAL EVENTS ITEMS AND INTERNAL
# STATES MISSINGNESS
miss_set$total_ext <- rowSums(miss_set[, c(1:52)])
miss_set$total_int <- rowSums(miss_set[, c(53:66)])
miss_set2 <- miss_set[, c(53:68)]  # just the total score, missingness
```

```r
# 4. CORRELATIONS
psych::corPlot(miss_set2[, 1:16], main = NULL, upper = FALSE)
```

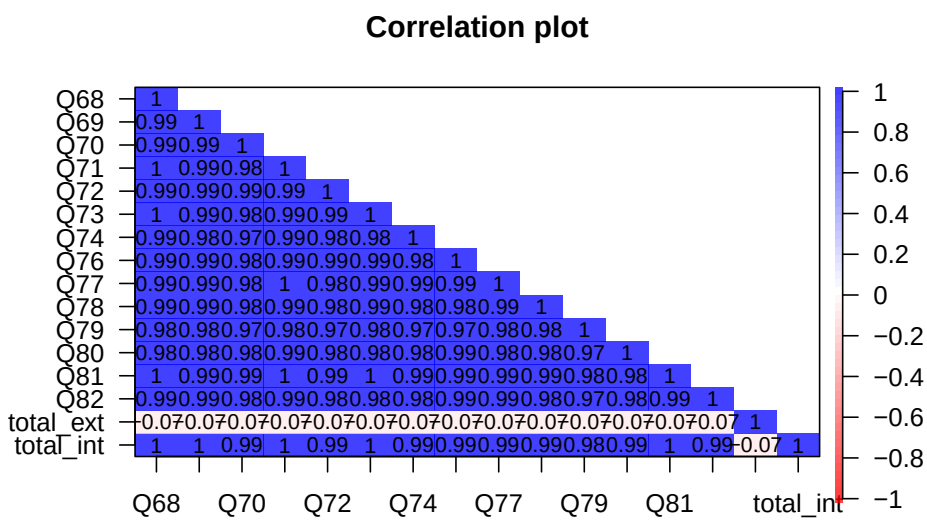


Figure 5.2: Correlation Matrix: External States and Internal States Missingness

Figure 5.2 illustrates the correlations among the Internal States items (dichotomized according to their missingness (1=missing, 0=all others)), the total score from the external states (total_ext), and the total score of internal item missingness (total_int). The correlation of total_ext with all the other items is nearly zero, suggesting that more exposure to abusive external states did ***not*** predict nonresponse to the Internal States items. This finding is consistent with the finding from Figure 5.1 that the missing data appear to missing completely at random.

Missing data methods. There are several different ways to handle missing data. A common method is to *delete all cases missing any data*. This is called **listwise** deletion of data. Its primary advantages are simplicity, expedience, and analysis based on complete information. The disadvantages of listwise are (1) disposal of the information offered by those examinees who responded to some items but not others, and (2) diminishment of the dataset, and with it, a reduction in statistical power. The "all" data file of all Likert items and

demographics items derived earlier reports data from 3,222 examinees. The following code chunk creates a subset of complete data. This smaller dataset has 1,841 examinees.

```
# LISTWISE DELETION OF MISSING DATA
all_l <- na.omit(all)  #  Likert items from columns 14-79 plus demographics post listwise
write.csv(all_l, "data/all_l.csv", row.names = FALSE)  # save data

# OTHER POSSIBLE LISTWISE DATASETS

# items <- items[complete.cases(items), ] # create new
# dataset without missing data
items_l <- na.omit(items)  # Likert items from columns 14-79 post listwise
ext_items_l <- na.omit(ext_items)  # Likert items for external events post listwise
int_items_l <- na.omit(int_items)  # Likert items for internal states post listwise
write.csv(items_l, "data/items_l.csv", row.names = FALSE)
write.csv(ext_items_l, "data/ext_items_l.csv", row.names = FALSE)
write.csv(int_items_l, "data/int_items_l.csv", row.names = FALSE)
```

Analysis can proceed with these two datasets:

1. the "all" dataset of all Likert and demographic items (N=3,222)
2. the "all_l" dataset of 1,841 examinees who reported complete data

Chapter 6

Item Analysis

In this chapter, we examine the technical quality of the items that Dan piloted for the SHAS. Here we address questions like the following:

- Which items, if any, are too "easy"? What is the impact of this?
- Which items, if any, are too "difficult"? What is the impact of this?
- Which items, if any, correlate weakly with others (and therefore might need to be refined or excluded from scaling?
- Which items, if any, have too many response categories which might be profitably collapsed?

We begin by defining item analysis and discussing its importance. Then we document our item analyses from an item response theory (IRT) framework. We conclude by summarizing what we learn from these analyses to help us refine these 66 items and possibly develop better ones in the future.

6.1 Item analyses in item response theory

Item response theory (IRT) is an established framework for empirically evaluating what items measure [Embretson and Reise, 2000, Wright and Masters, 1982]. IRT methods complement methods derived from classical test theory in several ways.

First, IRT places both examinees and items on the same interval scale. In classical test theory, examinees differ in their scores and items differ in their means or percent of examinees selecting a response; examinees and items are on different scales. By contrast, IRT methods allow measurement of persons and items on the ***same scale*** with *equal interval properties* of the *logit* scale** and resulting linear measures (Wright & Stone, 1999). In IRT we can estimate *item parameters independently of the characteristics of the calibrating sample*, and

we can estimate *person parameters apart from the difficulties of the items taken* (Masters, 1982; Rasch, 1966).

Second, IRT gives items diagnostic value. A core assumption of CTT is that items are *redundant*; they contribute meaningful variation to a total score but they do not occupy a distinct place on the score continuum. IRT is different. IRT does not assume that items are redundant; instead it investigates whether items make distinct contributions to the measurement of the latent trait. Whereas classical test theory measurement is principally concerned with the ratio of noise to signal in the variation contributed by an item, IRT models compute the probability of a certain response to an item given the level of the latent construct the individual possesses (i.e., spiritual abuse) and the relevant item's difficulty of endorsement. IRT is based on the logistic model and operates on the *logit* scale.

In what follows, we conduct item analyses through a comparison of several different IRT models: the Partial Credit Model (PCM), the Generalized Partial Credit Model (GPCM), and the Rating Scale Model (RSM). I credit Okan Bulut [2018] for his tutorial illustrating how to carry out IRT analyses in R using the `mirt` package [Chalmers, 2012].

6.1.1 IRT Model Assumptions

IRT makes two assumptions of the data. The first is unidimensionality of the latent trait. Is the trait one dimension? This is important for the SHAS because the questionnaire explicitly distinguished between items measuring external events and internal states. To check this assumption we first confirm the factor structure using confirmatory factor analysis (CFA) and Rasch Principal Component Analysis of Residuals (PCAR; Smith, 2002). The second assumption is local independence [Embretson and Reise, 2000]. This means that any set of items should not share any meaningful correlation once the latent variable has been accounted for (Edwards, Houts, & Cai, 2018).

To check the assumption of local independence, we used the Q3 statistic (Yen, 1984). The Q3 statistic index criteria are that the raw residual correlation between pairs of items should never exceed 0.10 (Marais & Andrich, 2008).

Using these criteria, we found XX items in possession of local dependence. The items that had the highest raw residual correlations had negative signs and no positive residual correlation. In other words, the assumption of local independence in this study (was)/(was not) met.

6.1.2 Partial Credit Model (PCM)

The first model we examine is the Partial Credit Model (PCM) [Masters, 1982]. This is a polytomous item response model belonging to the Rasch family. This means that it extends the Rasch model, which is based on dichotomous item responses (1=correct, 0=incorrect), by considering the probability of choosing

from among several response options such as those typically presented in Likert scales of agreement. The number of response categories in each item may vary [Embretson and Reise, 2000]. It follows that the PCM contains m (m + 1 being the number of response categories) threshold parameters, and each threshold parameter marks a category intersection [Masters, 1982]. Threshold sometimes refers to "step difficulty" or "step parameter" and illustrates the point on the latent trait continuum (e.g., spiritual abuse) where a response in category k becomes more likely than a response in category k − 1 (de Ayala, 2009). The PCM provides scores for items, persons, and step parameters on a logit scale. We begin by loading the `mirt` library and estimating the model on the data from the 66 SHAS items. This model constrains the slopes to 1 and freely estimates the variance parameters.

```
knitr::kable(rv, "simple", digits = 2, caption = "Partial Credit Model - Item Parameters - I
    full = TRUE)
```

Table 6.1: Partial Credit Model - Item Parameters - Item
Category Threshold Reversals

	a	b1
EQ7 - Pastor speaking on God's behalf	1	0.23
EQ10 - Asked to give up personal goals by pastor	1	1.26
EQ17 - Treated as less than b/c my sexual orientation	1	3.93
EQ18 - Church/pastor discourage critical thinking	1	-0.15
EQ19 - Behavior excessively monitored by pastor/group	1	0.42
EQ20 - Members pressured to give money despite financial hardship	1	0.97
EQ21 - Taught I would risk Hell if left my church/group	1	1.11
EQ22 - Expected to follow pastor/leader rules re: dating/marriage/sex	1	-0.39
EQ23 - Prayer replacing needed medical interventions	1	0.77
EQ24 - Expected to consult pastor/leader before making non-religious decisions	1	1.13
EQ26 - Pressured to forgive abuser while abuse was ongoing	1	1.41
EQ27 - Denied opportunities to serve b/c of my gender	1	1.64
EQ29 - Feeling special when in pastor's good graces; otherwise ignored	1	0.12
EQ30 - Deterred from seeking mental health treatment/counseling/medication	1	0.89
EQ32 - Feeling unable to express unhappiness	1	-0.30
EQ34 - Scripture used to justify physical punishment/severe discipline	1	0.10
EQ35 - Taught to distrust my emotions	1	-0.62
EQ38 - Unrealistic demands placed on my moral/religious behavior	1	-0.43
EQ39 - Made to feel shame over natural sexual desires (not actions)	1	-0.16
EQ40 - Threatening Divine punishment to keep group members in line	1	0.66
EQ41 - Denied opportunities to serve because of my sexual orientation	1	3.89
EQ44 - Love/acceptance offered only for high spiritual/moral performance	1	-0.01
EQ45 - Developmentally-inappropriate/anxious Hell/Satan/demons taught to young children	1	-0.10
EQ46 - Treated as less than because of my skin color	1	3.92
EQ47 - Medical care being postponed/withheld for religious reasons	1	1.56
EQ50 - Treated as less than because of my gender	1	1.58

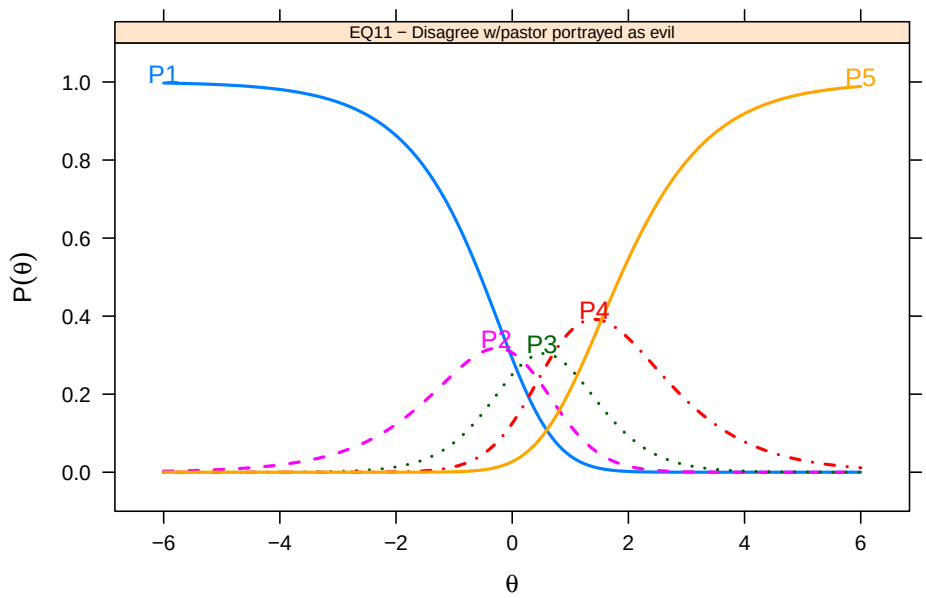
	a	b1
EQ51 - Extreme pressure to be pastor/missionary/spiritual leader	1	0.44
EQ52 - Blamed for harm I suffered rather than blaming who harmed me	1	0.73
EQ53 - Scripture used to justify abusive parent-child behavior	1	0.36
EQ54 - Being explicitly taught to distrust my intuitions	1	-0.36
EQ55 - Hearing cultural references in sermons unfamiliar to my race/eth. subculture	1	1.34
EQ56 - Being made to feel I was crazy/weird for doubts/questions	1	-0.18
EQ57 - Developing mental/physical ailments from conforming to group/leader's expectation	1	0.83
EQ58 - Terror/horror used to motivate religious decisions	1	0.04
EQ60 - Encouraged by leader to stay in abusive marriage	1	2.21
EQ61 - Developmentally-inappropriate/anxious end times descriptions taught to young children	1	0.05
EQ62 - Scripture used to justify physical violence	1	0.54
EQ65 - Cut off/shunned by more religious family members	1	1.22
EQ66 - Others treated as less than due to their sexual orientation	1	-0.58
IQ68 - Anxiety attacks triggered by religious stimuli	1	0.86
IQ72 - Feeling as if God harmed me directly	1	1.36
IQ76 - Feeling I wasted years of my life in a church/set of beliefs	1	-0.02
IQ81 - Nightmares about my negative religious experiences	1	0.94

Table 6.1 reports the item parameters estimated by the PCM. The first column shows the discrimination parameter, which is equal to "1" for all items. This is because the Partial Credit Model, similar to the Rasch model, constrains the discrimination parameter to be "1". The subsequent columns (b1 to b4) report the estimated thresholds or step parameters. Because the SHAS items all have five response categories, the PCM estimates four threshold parameters (b1 to b4) for each item. Ordered threshold parameters are not a requirement of the partial credit or generalized partial credit models [Embretson and Reise, 2000]. The important question, therefore, is which items have ordered thresholds and which have "reversals." A reversal suggests that a response category is not contributing reliable information and might be collapsed.

As shown in Table 6.1, most of the SHAS items (43 out of 66) had reversals. On each of these items, examinees who had experienced more overall lifetime spiritual abuse were more likely to report "Never" than "Once or twice". To collapse these categories would help these items contribute less noise and more reliable information.

Item plots help illustrate these statistics visually. Figure **??** is a plot of the option characeristic curves (also called category characteristic curves) of Q7.

```
# PCM ITEM PLOTS
library(directlabels)
plt <- plot(results.pcm, type = "trace", which.items = 3, main = "",
    par.settings = simpleTheme(lty = 1:4, lwd = 2), auto.key = list(points = FALSE,
        lines = TRUE, columns = 4))  # option curves
direct.label(plt, "top.points")
```



In this kind of plot, the x axis is the scale of the latent trait–in this case, lifetime exposure to spiritual abuse–expressed in logits ranging from -6 to +6. The y axis is the probability of selecting a given response (such as "Always") at each level of overall lifetime exposure to spiritual abuse. The blue line is the thus the probability of selecting "Never" for examinees ranging from low to high lifetime spiritual abuse. The reversal is that examinees just below zero are more likely to select "Never" than "Once or twice." In addition, examinees with above-average lifetime exposure to spiritual abuse were more likely to select "Always" than "Often". The item parameter statistics shown Table 6.1 indicate that this issue applies to 43 items. These items would contribute less noise and more reliable information if these response categories were collapsed.

Item fit. Two fit statistics assess the fit of an item to the PCM: the infit mean square (MNSQ) and the outfit mean square (MNSQ). The infit statistic places greater emphasis on unexpected responses that are close to the examinees and item location. The outfit is sensitive to unexpected responses that are far from the location [Bond and Fox, 2015]. The infit and outfit values identify potential unexpected response patterns. The expected value of infit or outfit for each item is 1.0, with a range of acceptable values ranging from 0.5 to 1.5. Values outside this boundary indicate a lack of fit between items and models.

```
knitr::kable(misfits, "simple", digits = 2, caption = "Partial Credit Model (PCM) - Misfitti
```

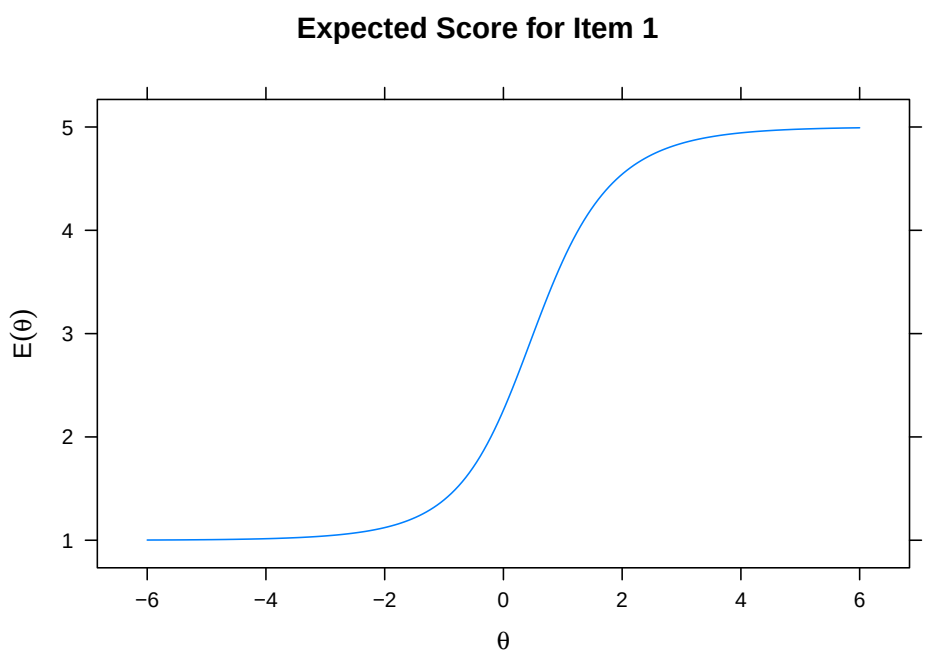
Table 6.2: Partial Credit Model (PCM) - Misfitting Items

	item	outfit	z.outfit
5	EQ17 - Treated as less than b/c my sexual orientation	2.81	7.05
15	EQ27 - Denied opportunities to serve b/c of my gender	2.67	17.85
28	EQ41 - Denied opportunities to serve because of my sexual orientation	6.71	15.06
33	EQ46 - Treated as less than because of my skin color	3.22	8.21
36	EQ50 - Treated as less than because of my gender	1.95	12.37
41	EQ55 - Hearing cultural references in sermons unfamiliar to my race/eth. subculture	2.83	22.03

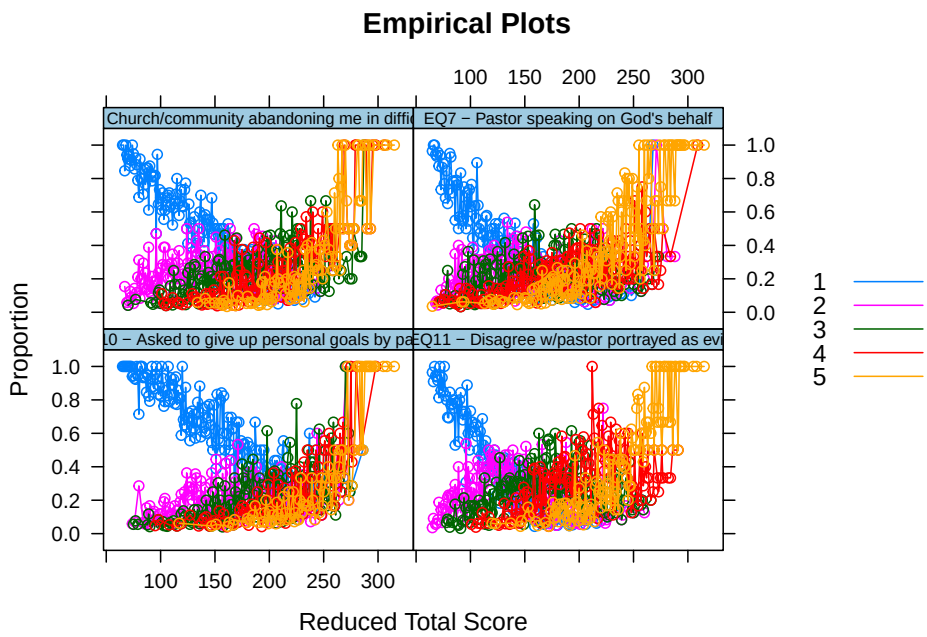
All but six of the 66 SHAS items had infit and outfit statistics within the acceptable range from 0.5 to 1.5. Statistics for these six items are presented here in 6.2. The large outfit values mean the observed patterns of responses to these items differed substantially from the patterns of responses expected by the PCM. These items capture the experiences of women, people of color, and LGBTQ persons.

Model fit. One way to evaluate the fit of the Partial Credit Model is to evaluate how well it fits the data from the item responses. The following table presents this information.

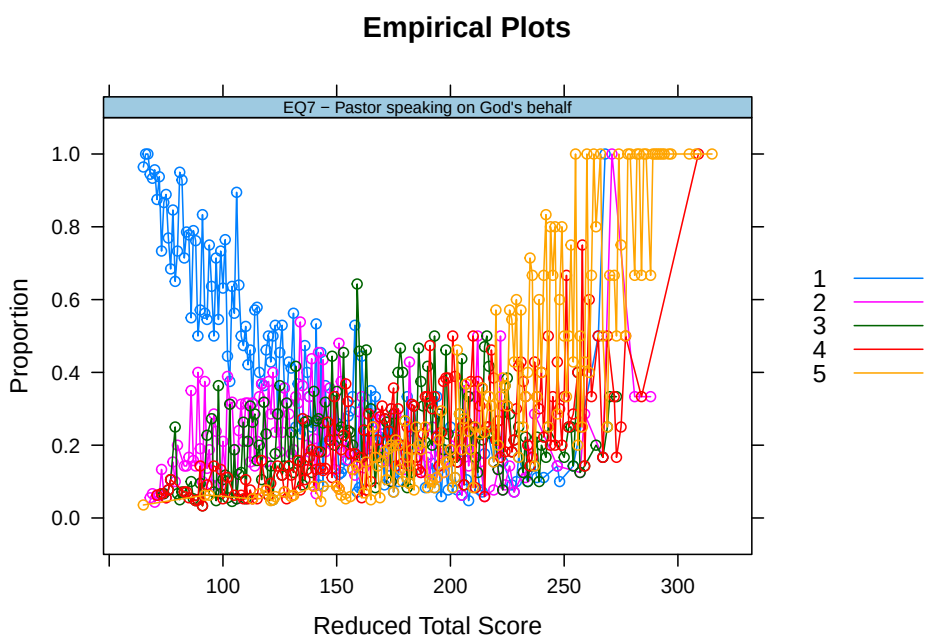
```
# PCM MODEL FIT
itemplot(results.pcm, 1, type = "score")
```



```r
empirical_plot(items, c(1, 2, 3, 4), main = "Empirical Plots")  # multiple items
```

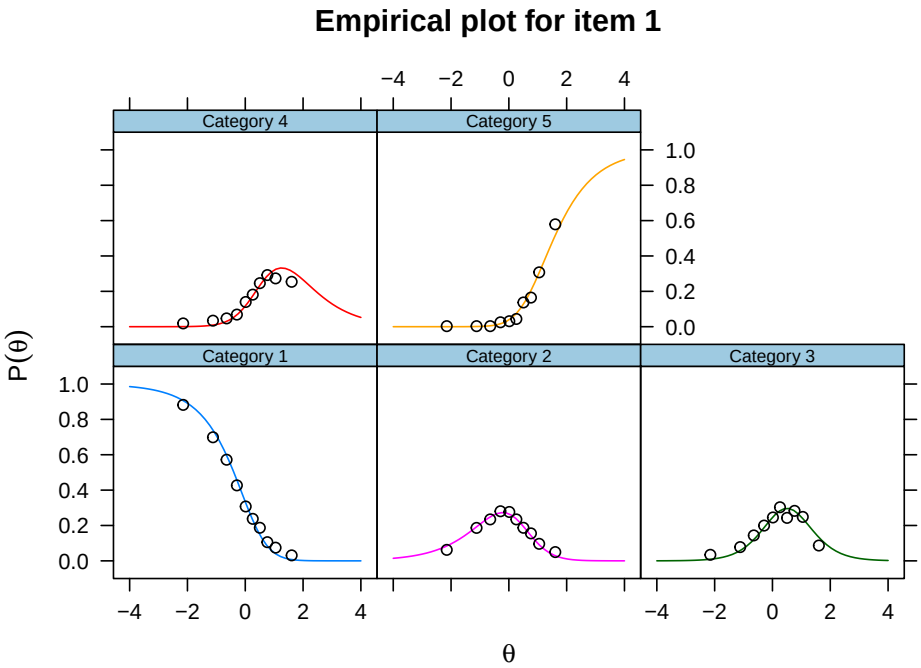



```r
empirical_plot(items, 1, main = "Empirical Plots")  # one item
```



```r
pcmfit <- as.data.frame(itemfit(results.pcm, na.rm = TRUE))
knitr::kable(pcmfit, caption = "Partial Credit Model (PCM) - Item Fit Statistics")
```

```r
itemfit(results.pcm, empirical.plot = 1)
```




```r
itemfit(results.pcm, "infit", na.rm = TRUE)
```

```
                                                                                    i
1                                                        EQ7 - Pastor speaking on God's beh
2                                                EQ10 - Asked to give up personal goals by pas
3                                                      EQ11 - Disagree w/pastor portrayed as e
4                                          EQ16 - Church/community abandoning me in difficult t
5                                            EQ17 - Treated as less than b/c my sexual orientat
6                                                EQ18 - Church/pastor discourage critical think
7                                            EQ19 - Behavior excessively monitored by pastor/gr
8                                 EQ20 - Members pressured to give money despite financial hards
9                                          EQ21 - Taught I would risk Hell if left my church/gr
10                          EQ22 - Expected to follow pastor/leader rules re: dating/marriage/
11                                            EQ23 - Prayer replacing needed medical interventi
12                 EQ24 - Expected to consult pastor/leader before making non-religious decisi
13                                             EQ25 - Feeling unable to raise questions and iss
14                                       EQ26 - Pressured to forgive abuser while abuse was ongo
15                                           EQ27 - Denied opportunities to serve b/c of my gen
16                                           EQ28 - Leadershi/group protecting abusive individu
17                          EQ29 - Feeling special when in pastor's good graces; otherwise igno
```

Table 6.3: Partial Credit Model (PCM) - Item Fit Statistics

item	S_X2
EQ7 - Pastor speaking on God's behalf	533.3564
EQ10 - Asked to give up personal goals by pastor	495.1835
EQ11 - Disagree w/pastor portrayed as evil	579.6394
EQ16 - Church/community abandoning me in difficult time	481.6413
EQ17 - Treated as less than b/c my sexual orientation	443.3413
EQ18 - Church/pastor discourage critical thinking	511.1011
EQ19 - Behavior excessively monitored by pastor/group	548.8232
EQ20 - Members pressured to give money despite financial hardship	543.2461
EQ21 - Taught I would risk Hell if left my church/group	556.8474
EQ22 - Expected to follow pastor/leader rules re: dating/marriage/sex	610.6849
EQ23 - Prayer replacing needed medical interventions	451.2542
EQ24 - Expected to consult pastor/leader before making non-religious decisions	439.9162
EQ25 - Feeling unable to raise questions and issues	505.6217
EQ26 - Pressured to forgive abuser while abuse was ongoing	481.9682
EQ27 - Denied opportunities to serve b/c of my gender	896.8917
EQ28 - Leadershi/group protecting abusive individuals	514.6797
EQ29 - Feeling special when in pastor's good graces; otherwise ignored	530.1984
EQ30 - Deterred from seeking mental health treatment/counseling/medication	569.1584
EQ31 - Shunned/ignored by pastor/church/group	526.1497
EQ32 - Feeling unable to express unhappiness	486.5294
EQ34 - Scripture used to justify physical punishment/severe discipline	586.1511
EQ35 - Taught to distrust my emotions	542.2495
EQ36 - Mental/physical problems interpreted as spiritual/moral weakness	553.8355
EQ37 - Made to feel less spiritually mature than pastor/leadership	548.0106
EQ38 - Unrealistic demands placed on my moral/religious behavior	568.9640
EQ39 - Made to feel shame over natural sexual desires (not actions)	473.9639
EQ40 - Threatening Divine punishment to keep group members in line	573.7718
EQ41 - Denied opportunities to serve because of my sexual orientation	435.2054
EQ42 - Seeing others shamed/shunned by pastor/leader/group	549.9676
EQ43 - Shamed by pastor/group for poor spiritual/moral performance	502.6761
EQ44 - Love/acceptance offered only for high spiritual/moral performance	618.0733
EQ45 - Developmentally-inappropriate/anxious Hell/Satan/demons taught to young children	548.4064
EQ46 - Treated as less than because of my skin color	249.0751
EQ47 - Medical care being postponed/withheld for religious reasons	347.5054
EQ49 - Leadership/group protecting and elevating abusive individuals	515.8695
EQ50 - Treated as less than because of my gender	1193.8223
EQ51 - Extreme pressure to be pastor/missionary/spiritual leader	613.6675
EQ52 - Blamed for harm I suffered rather than blaming who harmed me	486.9722
EQ53 - Scripture used to justify abusive parent-child behavior	541.7427
EQ54 - Being explicitly taught to distrust my intuitions	561.7842
EQ55 - Hearing cultural references in sermons unfamiliar to my race/eth. subculture	977.8732
EQ56 - Being made to feel I was crazy/weird for doubts/questions	700.4023
EQ57 - Developing mental/physical ailments from conforming to group/leader's expectation	508.4694
EQ58 - Terror/horror used to motivate religious decisions	583.6834
EQ59 - Pastor/group blame victim for their abuse	546.5812
EQ60 - Encouraged by leader to stay in abusive marriage	552.8959
EQ61 - Developmentally-inappropriate/anxious end times descriptions taught to young children	559.8063
EQ62 - Scripture used to justify physical violence	511.0822
EQ63 - Shamed by pastor/group for raising questions or concerns	557.6752
EQ64 - Witnessing women pressured to stay in unfaithful/abusive marriages	560.6293
EQ65 - Cut off/shunned by more religious family members	493.0967

```
                  EQ30 - Deterred from seeking mental health treatment/counseling/medicat
                                                EQ31 - Shunned/ignored by pastor/church/gr
                                                 EQ32 - Feeling unable to express unhappir
                      EQ34 - Scripture used to justify physical punishment/severe discipl
                                                       EQ35 - Taught to distrust my emoti
                    EQ36 - Mental/physical problems interpreted as spiritual/moral weakr
                         EQ37 - Made to feel less spiritually mature than pastor/leaders
                           EQ38 - Unrealistic demands placed on my moral/religious behav
                        EQ39 - Made to feel shame over natural sexual desires (not actio
                         EQ40 - Threatening Divine punishment to keep group members in l
                       EQ41 - Denied opportunities to serve because of my sexual orientat
                                   EQ42 - Seeing others shamed/shunned by pastor/leader/gr
                         EQ43 - Shamed by pastor/group for poor spiritual/moral performa
                     EQ44 - Love/acceptance offered only for high spiritual/moral performa
     EQ45 - Developmentally-inappropriate/anxious Hell/Satan/demons taught to young chilc
                                          EQ46 - Treated as less than because of my skin cc
                         EQ47 - Medical care being postponed/withheld for religious reas
                       EQ49 - Leadership/group protecting and elevating abusive individu
                                                EQ50 - Treated as less than because of my ger
                           EQ51 - Extreme pressure to be pastor/missionary/spiritual lea
                        EQ52 - Blamed for harm I suffered rather than blaming who harmec
                             EQ53 - Scripture used to justify abusive parent-child behav
                                   EQ54 - Being explicitly taught to distrust my intuiti
         EQ55 - Hearing cultural references in sermons unfamiliar to my race/eth. subcult
                           EQ56 - Being made to feel I was crazy/weird for doubts/questi
    EQ57 - Developing mental/physical ailments from conforming to group/leader's expectat
                                  EQ58 - Terror/horror used to motivate religious decisi
                                           EQ59 - Pastor/group blame victim for their ab
                                     EQ60 - Encouraged by leader to stay in abusive marri
EQ61 - Developmentally-inappropriate/anxious end times descriptions taught to young chilc
                                         EQ62 - Scripture used to justify physical viole
                             EQ63 - Shamed by pastor/group for raising questions or conce
                  EQ64 - Witnessing women pressured to stay in unfaithful/abusive marria
                                     EQ65 - Cut off/shunned by more religious family memb
                         EQ66 - Others treated as less than due to their sexual orientat
                                       IQ68 - Anxiety attacks triggered by religious stim
                                                 IQ69 - Lack of spiritual direction/purp
                                                                IQ70 - Feeling betrayed by
                IQ71 - Avoiding religious activities/settings to reduce distressing feeli
                                                IQ72 - Feeling as if God harmed me direc
                                                        IQ73 - Self-hatred or self-loath
                               IQ74 - Sadness over the loss of my faith/religious commun
                        IQ76 - Feeling I wasted years of my life in a church/set of beli
                             IQ77 - Anger upon reflecting on negative religious experier
                                                                IQ78 - A lack of self-wo
```

```
63                                                                                IQ79 - Distrust of
64                                                                               IQ80 - Feeling isola
65                                                IQ81 - Nightmares about my negative religious experien
66                                          IQ82 - Having trouble navigating life outside my church/commun
   outfit z.outfit infit z.infit
1   1.276    7.187 1.122   4.669
2   0.963   -0.671 0.988  -0.415
3   0.780   -7.399 0.799  -8.713
4   0.930   -1.824 0.939  -2.337
5   2.813    7.053 1.555   9.545
6   0.777   -7.906 0.804  -8.498
7   0.778   -6.464 0.836  -6.867
8   0.970   -0.618 1.014   0.513
9   1.203    3.704 1.043   1.575
10  1.104    2.990 1.099   3.596
11  0.941   -1.383 1.007   0.269
12  0.913   -1.766 0.936  -2.327
13  0.771   -9.221 0.791  -9.059
14  1.313    4.798 0.963  -1.247
15  2.671   17.851 1.502  15.055
16  0.848   -4.823 0.869  -5.376
17  1.014    0.428 1.026   1.060
18  0.848   -3.539 0.893  -4.155
19  1.012    0.318 1.006   0.216
20  0.882   -4.193 0.897  -4.308
21  1.119    3.474 1.102   3.963
22  0.976   -0.850 0.917  -3.397
23  0.792   -7.866 0.812  -8.212
24  0.845   -5.731 0.862  -5.881
25  0.680  -12.407 0.711 -13.043
26  0.994   -0.167 0.998  -0.085
27  0.788   -5.520 0.845  -6.321
28  6.705   15.057 1.534   8.204
29  0.763   -9.409 0.790  -9.222
30  0.748   -7.211 0.795  -8.281
31  0.715   -9.971 0.734 -11.859
32  0.978   -0.672 0.994  -0.251
33  3.223    8.213 1.392   4.076
34  0.986   -0.207 1.027   0.728
35  0.821   -5.079 0.865  -5.373
36  1.947   12.371 1.513  16.610
37  1.287    6.901 1.232   8.445
38  0.698   -7.911 0.759  -9.911
39  0.836   -4.767 0.896  -4.247
40  0.872   -4.519 0.866  -5.674
```

```
41  2.828   22.030 1.577  14.747
42  0.659  -12.674 0.687 -14.270
43  0.800   -4.843 0.896  -4.083
44  0.879   -3.921 0.920  -3.294
45  0.720   -8.510 0.778  -9.340
46  1.436    4.429 1.199   4.970
47  1.013    0.402 1.025   0.997
48  0.968   -0.821 1.037   1.426
49  0.722   -9.474 0.775  -9.632
50  0.900   -2.977 0.913  -3.471
51  1.183    3.247 1.051   1.715
52  1.013    0.443 0.984  -0.613
53  0.890   -2.575 0.959  -1.542
54  1.184    6.976 1.185   7.178
55  1.238    6.218 1.223   7.897
56  0.869   -4.950 0.891  -4.589
57  1.020    0.380 1.091   2.772
58  1.089    2.783 1.121   4.679
59  1.205    6.924 1.188   7.246
60  0.922   -2.415 0.925  -3.068
61  0.735  -11.202 0.745 -11.386
62  1.024    0.878 1.039   1.571
63  1.215    6.432 1.184   6.889
64  0.938   -2.471 0.946  -2.227
65  0.812   -4.346 0.909  -3.187
66  1.069    2.012 1.127   4.750
```

Table X "contains estimation results and the fit index model from each item of the SHAS." The estimates of item difficulty were between -X.XX TO X.XX (ie., -1.36 to 1.51) on the logit scale. Based on the item discrimination index (PTBIS; point-biserial correlation), which is analogous to classical test theory (item to-total-correlation), X SHAS items have high discrimination values in a positive direction, with no values below 0.30 or negative values. This indicates that XX items function well to distinguish persons with high versus low levels of spiritual abuse. For each item, we examined the step parameter of category endorsements and the statistics of each response for the items. X items, except item X, followed the step ordering requirement.

We found Item X (Step 1 = -0.XX, Step 2 = -0.XX, Step 3 = 0.XX) experienced threshold disordering as the thresholds were not ordered from lowest to highest. However, none of the infit and outfit statistics for response categories were greater than 2. This finding indicated that collapsing categories was not necessary because the disordered threshold still fit and did not violate the Rasch model (Linacre, 2010; 2018). Based on this information, we concluded that, in general, the SHAS items (fit)/(did not fit) the Rasch PCM model.

```
pcmscores <- fscores(results.pcm)
items <- cbind(items, pcmscores)
```

6.2 Summary

Chapter 7

Reliability

In this chapter, we addresses the reliability of the SHAS based on the 66 piloted items. We begin by defining reliability and discussing its importance. Then we report the results of our analyses of the reliability of the piloted SHAS items from both classical test theory (CTT) and item response theory (IRT) frameworks. Since these are cross-sectional data taken at one point in time, our work is limited to *internal consistency* reliability which focuses on the relationships between the items at one point in time rather than relationships between items or scores between two or more time points. We conclude by considering what this information contributes to improved SHAS instrumentation.

7.1 About reliability

Reliability is the extent to which items and their scores are consistently measuring the trait of interest. It is evidence that two examinees who provide the same responses to the same items, and by extension receive the same total scores, truly have the same level or amount of, or exposure to, the trait.

7.2 Reliability in Classical Test Theory (CTT)

One way to think about reliability in classical test theory is the ratio of signal to noise in the observed data. Do the various item responses and scores provide a true sort of examinees on the trait, or is all this variation merely noise?

The foundation of internal consistency reliability in CTT is relationships between item responses: correlations. Strong correlations among the items are evidence that the items are measuring the same construct. Thus we begin by examining correlation matrices.

```r
# HEATMAP OF CORRELATIONS
psych::corPlot(ext_items_l, upper = FALSE)
```

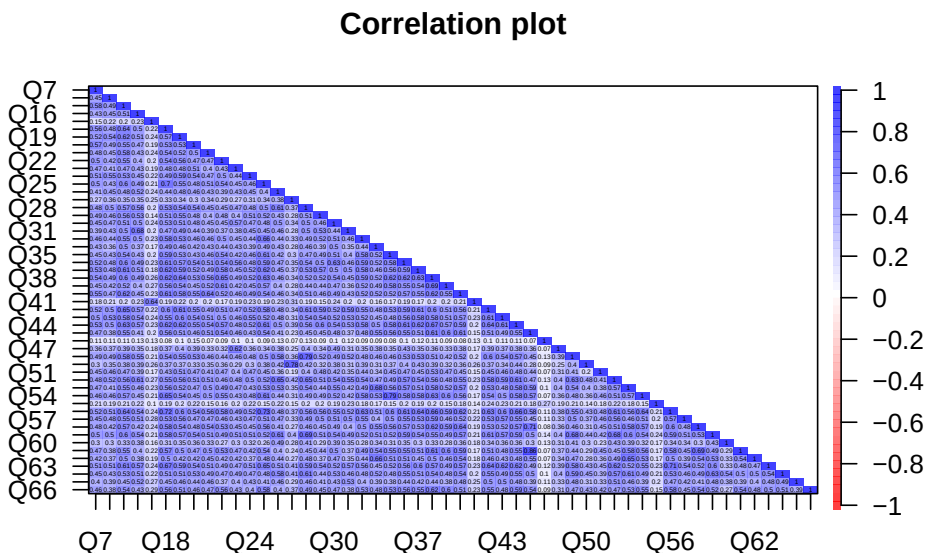


Figure 7.1: Correlation Matrix of External Events Items

Figure 7.1 is a correlation matrix of the External Events items for the *3,174* examinees who responded to all of these items. Each cell reports the correlation coefficient between scores of the two items. All of the cells are color coded to the scale of correlations which range from 0 (indicating no relationship) to 1 (indicating a perfect linear relationship). The cells along the diagonal are all 1.0 because they represent the correlation of the item with itself. Only one half of the matrix is necessary to report all the correlations among all the items.

The plot offers several interesting pieces of information. The first is that Q17–"Being treated as 'less than' because of my sexual orientation–does not quite fit. Responses to this item correlate weakly with responses to all the other items. It is possible that so few examinees experience this event that the responses vary very little, which would attenuate any correlation. Or it may be measuring something different than what the other items in this set. Similar could be said of Q27–"Being denied opportunities to serve/volunteer because of my gender."

```r
# HEATMAP OF CORRELATIONS
psych::corPlot(int_items_l, upper = FALSE)
```

Figure 7.2 is a correlation matrix of the Internal States items for the *3,051* examinees who responded to all of these items. This plot is interesting in that most of the correlations are in the moderate (0.50) range. They are sorting examinees in a similar way, but together they do not measure exactly the same

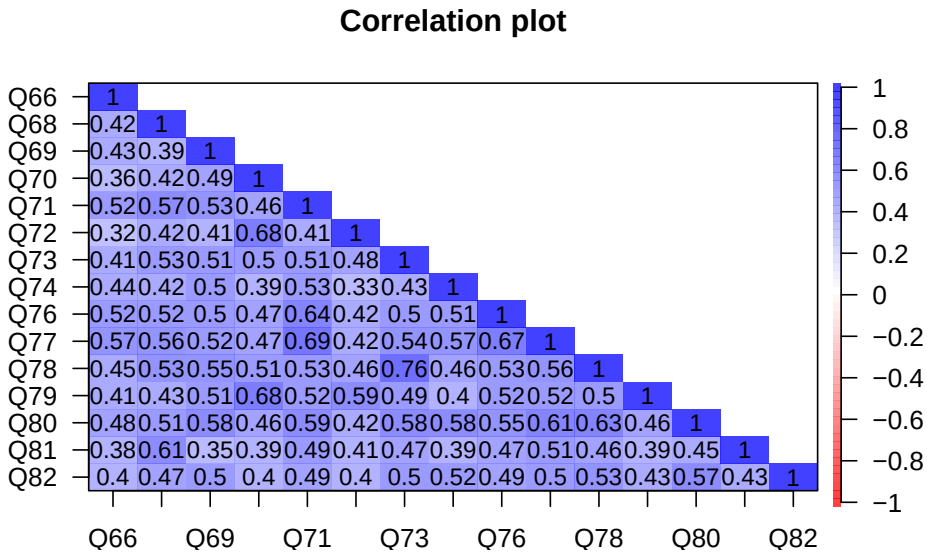


Figure 7.2: Correlation Matrix of Internal States Items

thing.

```
# HEATMAP OF CORRELATIONS
psych::corPlot(items_l, upper = FALSE)
```

This final matrix, Figure 7.3, reports the correlations among all of the Likert items among the *2,979* examinees who responded to all 66 items. Note that missing data are a challenge here, as these plots require complete data.

This plot offers several interesting insights. First, the preponderance of blue indicates that everything is positively correlated. This makes some sense because the items were worded and all the response categories were coded in the same direction.

Second, in this plot there are too many items and correlation coefficients in one place to be distinguishable, but with the colors it is possible to see an overall pattern. This plot orders the items by their order on the questionnaire so we can see how the External Events items correlate with the Internal Events items: moderately. It is possible that these are not two different scales.

Finally, we see at a glance a handful of items that correlate very weakly with all the other items. These items should be scrutinized for refinement and possible omission from the analysis.

Cronbach's coefficient alpha. Arguably the most well-known statistic for estimating internal consistency reliability is Cronbach's coefficient alpha (citation). This is a measure of the average inter-item correlation (?). It is also a measure of the ratio of signal to noise (X = T + e). A ballpark cutoff value is

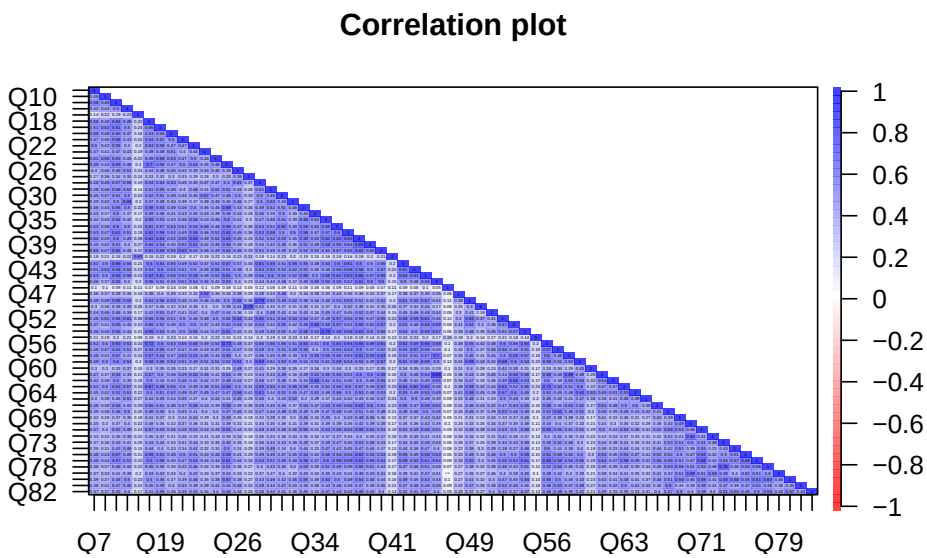


Figure 7.3: Correlation Matrix of All 66 Likert SHAS Items

0.70 [Nunnally and Bernstein, 1994].

```
# CRONBACH'S ALPHA
reliability(items)  # , itemal = TRUE, NA.Delete = TRUE, ml = TRUE)
```

```
Number of Items
66

Number of Examinees
2979

Coefficient Alpha
0.981
```

This output is not pretty, but it does report a Cronbach's alpha of 0.98. This is a high level of internal consistency reliability.

7.3 Reliability in Item Response Theory (IRT)

Wright and Masters (1982) developed reliability measures based on the Rasch measurement model, which have a different concept from classical test theory (e.g., Cronbach's alpha). Reliability is estimated both for persons and items, by means of Person Separation Reliability (PSR). This is an estimate of how well this instrument can distinguish respondents on the measured variable. In parallel, Item Separation Reliability (ISR) is an indication of how well items are

separated by the persons taking the test (Wright & Stone, 1999). The cutoff of separation reliabilities is > 0.80 (Bond & Fox, 2015).

7.4 Summary

Chapter 8

Scale Analysis

In this chapter, we examine the technical quality of the scales derived from the SHAS piloted items. Here we address questions like the following:

- **Person fit** - How well does the current scale fit the population of examinees? Are there ceiling or floor effects? To what extent might the existing items under-represent the full range of abuse?
- **Scale invariance** - Do the scales mean the same thing to different groups of examinees? To what extent do these scales (abuse constructs) vary by age, gender, and/or sexuality?
- **Developmental pathway** - Beyond total scores, what might an incremental scale of in terms of items look like? What items might represent the most severe or most frequent forms of spiritual abuse?

We begin by defining scaling and discussing the importance of scale analysis. We report the results of our scale analysis from within the item response theory (IRT) framework. We conclude by considering what this information contributes to improved SHAS instrumentation.

8.1 Person Fit

8.2 Scale Invariance

8.3 The Item-Person Map

The item-person map (also known as the Wright map) is a tool for visualizing the results of a calibration of items and persons. The Wright map was originally developed for use in test development but it also helps illuminate the results of surveys [Bulut, 2021]. As Bulut [2021] explains:

"The Partial Credit Model (PCM), which is an extension of the Rasch model for polytomous item responses, can be used for survey development and validation (Green & Frantom, 2002). We can analyze survey items using PCM and obtain item thresholds for each item. These thresholds indicate the magnitude of the latent trait required to select a particular response option (e.g., selecting disagree over strongly disagree). Using an item-person map, we can see thresholds for the items as well as the distribution of the latent trait for the respondents. This plot is quite useful for analyzing the alignment (or match) between the items and respondents answering the items."

```
# ITEM-PERSON MAP #1: FOR ALL THE SHAS ITEMS
source("code/itempersonmap.R")
library("mirt")
items <- read.csv("data/items.csv")
mod <- mirt(items, 1, itemtype = "Rasch", technical = list(removeEmptyRows = TRUE),
    verbose = FALSE)  # estimate the Partial Credit Model
itempersonmap(mod)  # create the item-person map
```

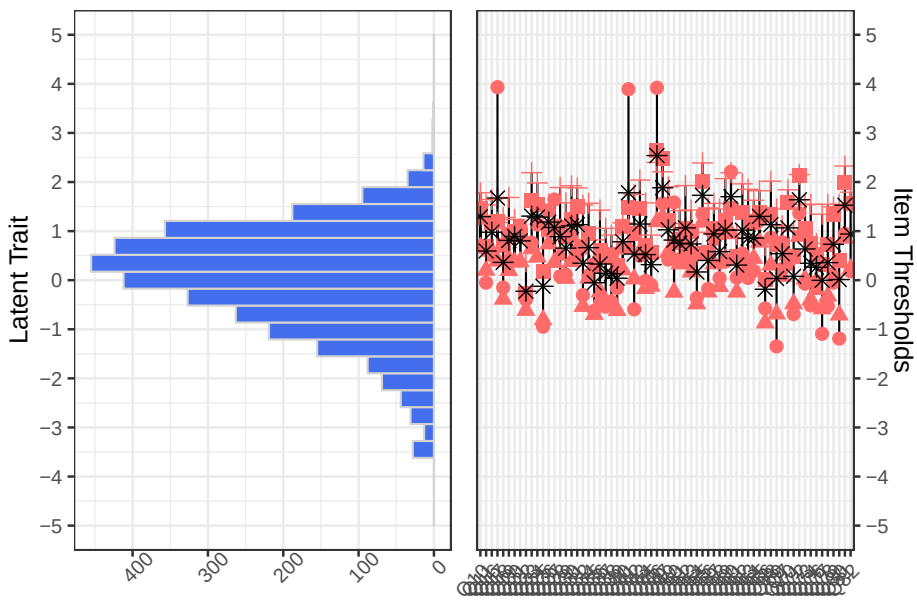


Figure 8.1: Item-Person Map: All Likert Items

Figure 8.1 is an item-person map of all 66 Likert SHAS items against all of the examinees. In the plot, red points indicate item thresholds for each item (three thresholds separating four response categories) and the asterisk indicates the average value of the thresholds for each item. The higher these thresholds, the more spiritual abuse the examinee has experienced. On the left-hand side of the plot, we also see the distribution of examinees along the latent trait (i.e., the construct of spiritual abuse).

This plot offers several insights at a general level. First, the plot of examinees on the left suggests that the latent trait of spiritual abuse is normally distributed, with a slight negative skew. Very few examinees reported very low or very high levels of exposure to spiritual abuse; most reported a moderate to slightly higher than average level of spiritual abuse. Second, with this plot it is difficult to distinguish individual items but it does show that they tend to clusteraround this "slightly higher than average" level of spiritual abuse, with none capturing a very low level of abuse and just a handful of items measuring much higher than average levels of spiritual abuse. Three items had response categories that captured very high levels of spiritual abuse.

8.4 Information curves

These provide important information about where along the scale of the latent trait the item or set of items provides the most reliable information. If there are too few "easy" or "difficult" items, then these extreme scores for examinees are less reliable.

8.5 Person Separation Reliability

8.6 Item Separation

8.7 Summary

Chapter 9

Discussion

References

Bibliography

American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. *Standards for Educational and Psychological Testing*. American Educational Research Association, Washington, DC, 2014.

Trevor G. Bond and Christine M. Fox. *Applying the Rasch Model: Fundamental Measurement in the Human Sciences*. Routledge, New York, New York, 2015. ISBN 978-0-415-83342-4.

Okan Bulut. Estimating polytomous irt models in r. https://rpubs.com/okan bulut/lecture2, February 2018.

Okan Bulut. Okan bulut: 5 ways to effectively visualize survey data, 2021. URL https://okan.cloud/posts/2021-03-04-5-ways-to-effectively-visualize-survey-data/.

R. Philip Chalmers. mirt: A multidimensional item response theory package for the R environment. *Journal of Statistical Software*, 48(6):1–29, 2012. doi: 10.18637/jss.v048.i06.

Susan E. Embretson and Steven P. Reise. *Item Response Theory for Psychologists*. Lawrence Erlbaum Associates, Mahwah, New Jerey, 2000. ISBN 0-8058-2819-2.

R.J.A. Little. A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, 83:1198–1202, 1988.

Geoff N. Masters. A rasch model for partial credit scoring. *Psychometrika*, 47: 149–174, 1982.

Patrick E. McKnight, Katherine M. McKnight, Souraya Sidani, and Aurelio Jose Figueredo. *Missing Data: A Gentle Introduction*. The Guilford Press, New York, New York, 2007.

Jum C. Nunnally and Ira H. Bernstein. *Psychometric Theory*. McGraw-Hill, Inc., New York, New York, 3rd edition, 1994.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2022. URL https://www.R-project.org/.

Steven P. Reise and Niels G. Waller. Item response theory and clinical measurement. *Annual Review of Clinical Psychology*, 5:27–48, 2009. doi: 10.1146/annurev.clinpsy.032408.153553.

Matthias Templ, Alexander Kowarik, Andreas Alfons, Gregor de Cillia, and Wolfgang Rannetbauer. *VIM: Visualization and Imputation of Missing Values*, 2021. URL https://github.com/statistikat/VIM. R package version 6.1.1.

Benjamin D. Wright and Geofferey N. Masters. *Rating Scale Analysis*. Mesa Press, Chicago, Illinois, 1982.

Yihui Xie. *bookdown: Authoring Books and Technical Documents with R Markdown*, 2022. URL https://CRAN.R-project.org/package=bookdown. R package version 0.26.